Solutions & Grading Rubric

General Instructions

- **Questions 1-10:** 3 Points each.
- **Questions 11-12:** 7.5 Points each.
- **Total:** 45 Points.

SET 1 (Circle)

1. **Vector** $\mathbf{x} = [-4, 3, 0, 0]^T$.
   Solution:

   - (i) $L_0 = 2$ (Number of non-zero elements)
   - (ii) $L_1 = |-4| + |3| = 7$
   - (iii) $L_2 = \sqrt{(-4)^2 + 3^2} = \sqrt{25} = 5$
   - (iv) $L_\infty = \max(|-4|, |3|) = 4$

   Rubric: *0.75 points for each correct norm. Total 3 pts.*

2. **Optimization:** Minimize $\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$ s.t. $||\mathbf{w}||_2^2 = 1$.
   Solution:

   - (A) (iii) Smallest
   - (B) (i) $\mathbf{X}^T\mathbf{X}$

   Rubric: *1.5 points for each blank. Total 3 pts.*

3. **Eigenfaces:** 1000 images of 100×100.
   Solution:

   - Checked: Covariance matrix is of size 10000×10000.
   - Checked: Practical rank of the covariance matrix will be far lesser than $min(N, d)$.

   Rubric: *0 if any incorrect option is marked, 1.5 marks if at least half of the correct options are marked, 3 if all correct options are marked.*

4. **Spam Classifier:** $\theta = 0.5$.
   Scores: $0.7, 0.65, 0.92, 0.75, 0.45, 0.8, 0.3, 0.8, 0.1, 0.1$
   Labels: $+, -, +, +, -, +, +, +, -, -$
   Solution:

   - Predictions (≥ 0.5): $+, +, +, +, -, +, -, +, -, -$ (Indices: 1, 2, 3, 4, 6, 8 are Predicted Positive).

- TP (Pred + / Actual +): Indices 1, 3, 4, 6, 8 (Count: 5)
- FP (Pred + / Actual -): Index 2 (Count: 1)
- FN (Pred - / Actual +): Index 7 (Count: 1)
- **Precision** $= TP/(TP + FP) = 5/6 \approx 0.83$
- **Recall** $= TP/(TP + FN) = 5/6 \approx 0.83$

Rubric: *1.5 pts for Precision, 1.5 pts for Recall.*

5. **1-NN Manhattan Distance.**
Train: $A(2, 0.5, +), B(-3, 0.5, -), C(0, 3, +), D(0, -3, -)$
Solution:

- Test 1 $(0, 0, +)$: NN is A (Dist 2.5). Correct.
- Test 2 $(1, 1, +)$: NN is A (Dist 1.5). Correct.
- Test 3 $(-1, 1, +)$: NN is B (Dist 2.5). Incorrect.
- Test 4 $(-1, -1, -)$: NN is D (Dist 3). Correct.
- Test 5 $(1, -1, -)$: NN is A (Dist 2.5). Incorrect.
- **Accuracy:** $3/5 = 60\%$ (or 0.6)

Rubric: *3 pts for correct accuracy. No partial credits.*

6. **SVD** $(m > n)$.
Solution:

- (a) Columns
- (b) Rows and Columns
- (c) Both AA^T and A^TA (Accept either or both, context implies singular value relation).

Rubric: *1 pt for each blank. for (b), 0.5 mark if only one of rows/columns is marked.*

7. **Normalization.** Mean Centering.
Data: $\{(2, 3), (3, 2), (2, -3), (0, 3), (0, 0), (1, 1), (-1, 1)\}$
Solution:

- Mean $x = 1$, Mean $y = 1$. Mean Vector $\mu = [1, 1]^T$.
- $\mathcal{D}' = \{[1, 2]^T, [2, 1]^T, [1, -4]^T, [-1, 2]^T, [-1, -1]^T, [-1, -1]^T, [-2, 0]^T\}$

Rubric: *1 pt for calculating mean, 2 pts for correct new set. allow upto 2 numerical mistakes with 0.5 partial marks deduction.*

8. **MSE Loss Function.**
Solution: $J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)^2$ (Sum without $1/N$ is also acceptable with 0.5 marks deducted). ***Rubric:*** *3 pts for correct formula. Deduct 0.5 if 1/N is missed.*

9. **PCA Gradient Ascent.**
Solution: (b) $\mathbf{v}_{k+1} = (I + \eta\Sigma)\mathbf{v}_k$ (derived from $\mathbf{v} + \eta\Sigma\mathbf{v}$). ***Rubric:*** *3 pts.*

10. **Covariance of points on line** $y = -x$ **(45°).**
Solution: (a) $\lambda_2 = 0$ (and d) Σ is singular). ***Rubric:*** *Option a: 2 points, option d: 1 point*

11. **Gradient Descent:** $\mathcal{L}(w) = 2w^2 - 4w + 7$.
Solution:

- (i) Update: $w_{k+1} = w_k - \eta(4w_k - 4)$.
- (ii) $w^0 = 2.0, \eta = 0.2$.
  $w^1 = 2.0 - 0.2(4(2) - 4) = 2.0 - 0.8 = 1.2$.
  $w^2 = 1.2 - 0.2(4(1.2) - 4) = 1.2 - 0.2(0.8) = 1.04$.
- (iii) Optima at $w = 1$. $\mathcal{L}(1) = 2 - 4 + 7 = 5$.

Rubric: *2.5 pts each for (i), (ii), (iii).*

12. **Regression Calculation.** Data: $\{(1, 2), (2, 3), (3, 4)\}$.
    Solution:

    - $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$ (with bias), $\mathbf{y} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$.
    - $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = [1, 1]^T$.
    - Equation: $y = 1 + 1x$.

    Rubric: *Construct X: 2.5, construct y: 1, substitute in the formula: 1, calculate inverse: 1.5, final result: 1.5.*

SET 2 (Double Cross)

1. **Vector** $\mathbf{x} = [-6, 3, 2, 0]^T$.
   Solution:

   - (i) $L_0 = 3$
   - (ii) $L_1 = 11$
   - (iii) $L_2 = 7$
   - (iv) $L_\infty = 6$

   Rubric: *3 pts (0.75 each).*

2. **Optimization:** Maximize $\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}$.
   Solution:

   - (A) (i) Largest
   - (B) (i) $\mathbf{X}^T\mathbf{X}$

   Rubric: *1.5 points for each blank. Total 3 pts.*

3. **Eigenfaces:** Duplicated photo 1000 times.
   Solution:

   - True: "If all images were identical, mean will be the same as that of Raju's original face."
   - True: "All elements of the covariance matrix will be zero."

   Rubric: *0 if any incorrect option is marked, 1.5 marks if at least half of the correct options are marked, 3 if all correct options are marked.*

4. **Spam Classifier:** $\theta = 0.2$.
   Solution:

   - Pred +: All except last two (Indices 1-8).
   - TP: 6, FP: 2, FN: 0.
   - **Precision:** $6/8 = 0.75$
   - **Recall:** $6/6 = 1.0$

   Rubric: *1.5 pts for Precision, 1.5 pts for Recall.*

5. **3-NN Manhattan Distance.**
   Solution:

   - Accuracy: 4 correct out of 5 = 80% (or 0.8).
     (Failed on Test point 5).

   Rubric: *3 pts for correct accuracy. No partial credits.*

6. **SVD** $(m > n)$.
   Solution:

   - (a) Columns
   - (b) Rows and Columns
   - (c) Both AA^T and A^TA (Accept either or both, context implies singular value relation).

Rubric: *1 pt for each blank. for (b), 0.5 mark if only one of rows/columns is marked.*

7. **Normalization.** (Same as Set 1).
   Solution: $\mathcal{D}' = \{[1,2]^T, [2,1]^T, [1,-4]^T, [-1,2]^T, [-1,-1]^T, [-1,-1]^T, [-2,0]^T\}$. ***Rubric:*** *1 pt for calculating mean, 2 pts for correct new set. allow upto 2 numerical mistakes with 0.5 partial marks deduction.*

8. **Ridge Regression Loss.**
   Solution: $J(\mathbf{w}) = \frac{1}{N}\sum(y_i - \mathbf{w}^T\mathbf{x}_i)^2 + \lambda||\mathbf{w}||_2^2$. ***Rubric:*** *3 pts for correct formula. Deduct 0.5 if 1/N is missed.*

9. **PCA Gradient Ascent.**
   Solution: (b) $\mathbf{v}_{k+1} = (I + \eta\Sigma)\mathbf{v}_k$ (derived from $\mathbf{v} + \eta\Sigma\mathbf{v}$). ***Rubric:*** *3 pts.*

10. **Covariance of points on line $y = -x$ ($45°$).**
    Solution: (a) $\lambda_2 = 0$ (and d) Σ is singular). ***Rubric:*** *Option a: 2 points, option d: 1 point*

11. **Gradient Descent:** $\mathcal{L}(w) = 2w^2 - 4w + 7$.
    Solution:

    - (i) Update: $w_{k+1} = w_k - \eta(4w_k - 4)$.
    - (ii) $w^0 = 0.0, \eta = 0.2$.
      $w^1 = 0 - 0.2(-4) = 0.8$.
      $w^2 = 0.8 - 0.2(3.2 - 4) = 0.96$.
    - (iii) Optima $w = 1$, $\mathcal{L}(1) = 5$.

    Rubric: *2.5 pts each for (i), (ii), (iii).*

12. **Regression:** $\{(1,-1),(2,0),(3,1)\}$.
    Solution:

    - Result: $\mathbf{w}^* = [-2,1]^T$. Equation: $y = -2 + 1x$.

    Rubric: *Construct X: 2.5, construct y: 1, substitute in the formula: 1, calculate inverse: 1.5, final result: 1.5.*

SET 3 (Star)

1. **Vector** $\mathbf{x} = [-4, 2, 0, 4]^T$.
   Solution:

   - (i) $L_0 = 3$, (ii) $L_1 = 10$, (iii) $L_2 = 6$, (iv) $L_\infty = 4$.

   Rubric: *0.75 points for each correct norm. Total 3 pts.*

2. **Optimization:** Minimize $\mathbf{w}^T\mathbf{X}\mathbf{X}^T\mathbf{w}$ ($m < n$ case).
   Solution:

   - (A) (iii) Smallest
   - (B) (ii) $\mathbf{X}\mathbf{X}^T$

   Rubric: *1.5 points for each blank. Total 3 pts.*

3. **Eigenfaces:** 1000 images, 5 students.
   Solution:

   - a: True: "Only the initial couple of hundred eigen values are significant..."
   - c: True: "If the 1000 images were from 5 students...closer to 5 and far from 200."
   - e: True: "If all images were rotated by 180° ... rank won't change."

   Rubric: *0 if any incorrect option is marked, 1.5 marks if at least half of the correct options are marked, 3 if all correct options are marked.*

4. **Spam Classifier:** $\theta = 0.6$.
   Solution:

   - TP: 5, FP: 1, FN: 1.
   - **Precision:** $5/6 \approx 0.83$
   - **Recall:** $5/6 \approx 0.83$

   Rubric: *1.5 pts for Precision, 1.5 pts for Recall.*

5. **1-NN Euclidean.**
   Solution: Accuracy $3/5 = 60\%$. ***Rubric:*** *3 pts for correct accuracy. No partial credits.*

6. **SVD** ($m > n$).
   Solution:

   - (a) Columns
   - (b) Rows and Columns
   - (c) Both AA^T and A^TA (Accept either or both, context implies singular value relation).

   Rubric: *1 pt for each blank. for (b), 0.5 mark if only one of rows/columns is marked.*

7. **Normalization.** Mean Centering.
   Data: $\{(2, 3), (3, 2), (2, -3), (0, 3), (0, 0), (1, 1), (-1, 1)\}$
   Solution:

   - Mean $x = 1$, Mean $y = 1$. Mean Vector $\mu = [1, 1]^T$.
   - $\mathcal{D}' = \{[1, 2]^T, [2, 1]^T, [1, -4]^T, [-1, 2]^T, [-1, -1]^T, [-1, -1]^T, [-2, 0]^T\}$

Rubric: *1 pt for calculating mean, 2 pts for correct new set. allow upto 2 numerical mistakes with 0.5 partial marks deduction.*

8. **LASSO Loss.**
   Solution: $J(\mathbf{w}) = \frac{1}{N}\sum(y_i - \mathbf{w}^T\mathbf{x}_i)^2 + \lambda||\mathbf{w}||_1$. ***Rubric:*** *3 pts for correct formula. Deduct 0.5 if 1/N is missed.*

9. **PCA Gradient Ascent.**
   Solution: (b) $\mathbf{v}_{k+1} = (I + \eta\Sigma)\mathbf{v}_k$ (derived from $\mathbf{v} + \eta\Sigma\mathbf{v}$). ***Rubric:*** *3 pts.*

10. **Covariance of points on line $y = -x$ ($45°$).**
    Solution: (a) $\lambda_2 = 0$ (and d) Σ is singular). ***Rubric:*** *Option a: 2 points, option d: 1 point*

11. **Gradient Descent:** $\mathcal{L}(w) = 4w^2 - 8w + 7$.
    Solution:

    - (i) Update: $w_{k+1} = w_k - \eta(8w_k - 8)$.
    - (ii) $w^0 = 2.0, \eta = 0.1$.
      $w^1 = 2.0 - 0.1(8) = 1.2$.
      $w^2 = 1.2 - 0.1(1.6) = 1.04$.
    - (iii) Optima $w = 1$, $\mathcal{L}(1) = 3$.

    Rubric: *2.5 pts each for (i), (ii), (iii).*

12. **Regression:** $\{(1,0),(2,1),(3,2)\}$.
    Solution:

    - Result: $\mathbf{w}^* = [-1, 1]^T$. Equation: $y = -1 + 1x$.

    Rubric: *Construct X: 2.5, construct y: 1, substitute in the formula: 1, calculate inverse: 1.5, final result: 1.5.*

SET 4 (Cross)

1. **Vector** $\mathbf{x} = [-8, 6, 0, 0]^T$.
   Solution:

   - (i) $L_0 = 2$, (ii) $L_1 = 14$, (iii) $L_2 = 10$, (iv) $L_\infty = 8$.

   Rubric: *0.75 points for each correct norm. Total 3 pts.*

2. **Optimization:** Maximize $\mathbf{w}^T \mathbf{X}\mathbf{X}^T \mathbf{w}$.
   Solution:

   - (A) (i) Largest
   - (B) (ii) $\mathbf{X}\mathbf{X}^T$

   Rubric: *1.5 points for each blank. Total 3 pts.*

3. **Eigenfaces:** 800 images.
   Solution:

   - a. True: "Number of non-zero eigenvalues... at most 799."
   - b. True: "Small number... explain most variance."
   - c. True: "images from 10 individuals...rank closer to 10."
   - e. True: "Fixed geometric transformation... will not change rank."
   - f. True: "Random geometric transformations... increase rank."

   Rubric: *0 if any incorrect option is marked, 1.5 marks if at least half of the correct options are marked, 3 if all correct options are marked.*

4. **Spam Classifier:** $\theta = 0.7$.
   Solution:

   - TP: 5, FP: 0, FN: 1.
   - **Precision:** $5/5 = 1.0$
   - **Recall:** $5/6 \approx 0.83$

   Rubric: *1.5 pts for Precision, 1.5 pts for Recall.*

5. **3-NN Euclidean.**
   Solution: Accuracy $4/5 = 80\%$. ***Rubric:*** *3 pts for correct accuracy. No partial credits.*

6. **SVD** ($m > n$).
   Solution:

   - (a) Columns
   - (b) Rows and Columns
   - (c) Both AA^T and $A^T A$ (Accept either or both, context implies singular value relation).

   Rubric: *1 pt for each blank. for (b), 0.5 mark if only one of rows/columns is marked.*

7. **Normalization.** Mean Centering.
   Data: $\{(2, 3), (3, 2), (2, -3), (0, 3), (0, 0), (1, 1), (-1, 1)\}$
   Solution:

- Mean $x = 1$, Mean $y = 1$. Mean Vector $\mu = [1, 1]^T$.
- $\mathcal{D}' = \{[1, 2]^T, [2, 1]^T, [1, -4]^T, [-1, 2]^T, [-1, -1]^T, [-1, -1]^T, [-2, 0]^T\}$

Rubric: *1 pt for calculating mean, 2 pts for correct new set. allow upto 2 numerical mistakes with 0.5 partial marks deduction.*

8. **MSE Loss Function.**
Solution: $J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^T\mathbf{x}_i)^2$ (Sum without $1/N$ is also acceptable with 0.5 marks deducted). ***Rubric:*** *3 pts for correct formula. Deduct 0.5 if 1/N is missed.*

9. **PCA Gradient Ascent.**
Solution: (b) $\mathbf{v}_{k+1} = (I + \eta\Sigma)\mathbf{v}_k$ (derived from $\mathbf{v} + \eta\Sigma\mathbf{v}$). ***Rubric:*** *3 pts.*

10. **Covariance of points on line $y = -x$ ($45°$).**
Solution: (a) $\lambda_2 = 0$ (and d) Σ is singular). ***Rubric:*** *Option a: 2 points, option d: 1 point*

11. **Gradient Descent:** $\mathcal{L}(w) = 4w^2 - 8w + 7$.
Solution:

- (i) Update: $w_{k+1} = w_k - \eta(8w_k - 8)$.
- (ii) $w^0 = 0.0, \eta = 0.1$.
$w^1 = 0 - 0.1(-8) = 0.8$.
$w^2 = 0.8 - 0.1(6.4 - 8) = 0.96$.
- (iii) Optima $w = 1$, $\mathcal{L}(1) = 3$.

Rubric: *2.5 pts each for (i), (ii), (iii).*

12. **Regression:** $\{(1, 3), (2, 4), (3, 5)\}$.
Solution:

- Result: $\mathbf{w}^* = [2, 1]^T$. Equation: $y = 2 + 1x$.

Rubric: *Construct X: 2.5, construct y: 1, substitute in the formula: 1, calculate inverse: 1.5, final result: 1.5.*